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# LESSON PLAN IN SCIENCE 10

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| **Ms. Althea C. Calinawan**<br>**Sir John Steve C. Landiza**<br>Teaching Intern | **January 26, 2026**<br>Date of Teaching |
| **Ma'am Blezy V. Guiroy**<br>Teaching Internship Mentor | **Grade 10**<br>Grade Level |
| **January 25, 2026**<br>Date of Submission | **9:30 A.M-2:15 P.M**<br>Time Schedule |

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| **Content Standard** | *The learners demonstrate an understanding of:*<br>- how gases behave based on the motion and relative distances between gas particles |
| **Learning Competencies** | *The learners shall be able to:*<br>- investigate the relationship between: volume and pressure at constant temperature of a gas; volume and temperature at constant pressure of a gas; explain these relationships using the Kinetic Molecular Theory. (S10MT-IVa-b-21) |

## I. Intended Learning Outcomes (Objectives)

At the end of the lesson and through varied learning activities, the Grade 10 learners, with at least 75% proficiency, will be able to:

a. explain how gas particles move and interact based on the Kinetic Molecular Theory;
b. formulate operational definitions of gas properties based on observed particle behavior; and
c. manipulate variable controls in a virtual simulation to isolate specific gas properties.

## II. Learning Content

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| A. **Topic (s)** | - **Kinetic Molecular Theory, Properties of Gases** |
| B. **Key Concepts** | (1) Kinetic Molecular Theory (KMT) is a scientific model that explains the behavior of gases based on the motion and arrangement of their particles.<br><br>• Gases are composed of molecules. The particles are far apart and occupy almost all the volume of the container.<br>• Gas particles are in continuous, random motion.<br>• There is negligible attractive or repulsive force between or among gas molecules.<br>• Collisions between gas particles are completely elastic.<br>• Movement of gas molecules is affected by temperature.<br><br>(2) Gases can be observed through four physical properties: pressure, volume, amount of gas, and temperature.<br><br>• Pressure (P): the force gas particles exert on the walls of their container.<br>• Volume (V): the space occupied by the gas, which changes to fit the container.<br>• Amount of gas (n): the quantity of gas, usually measured in moles.<br>• Temperature (T): indicates the average kinetic energy of gas particles |
| C. **References** *(Print, electronic, etc.)* | - SCIENCE 10 Learner's Module, Department of Education, Republic of the Philippines https://drive.google.com/file/d/13Hki4N89NxOUNh5so8Foyij3d1pUm1tE/view?usp=sharing<br><br>- Libretexts. (2023, January 30). *Kinetic Molecular Theory of Gases*. Chemistry LibreTexts. https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Physical_Properties_of_Matter/States_of_Matter/Properties_of_Gases/Kinetic_Theory_of_Gases/Kinetic_Molecular_Theory_of_Gases<br><br>- Gas simulation: Gas Properties |

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| D. **Instructional Materials** | - Laptop<br>- Canva Presentation<br>- Science Textbook |
| E. **Integration** *(Values, SDG/ESD, GAD, etc.)* | - **Quality Education (SDG 4)**: Enhances abstract reasoning and scientific literacy by visualizing invisible molecular behavior (KMT) to explain observable phenomena, allowing learners to bridge the gap between microscopic models and macroscopic reality.<br>- **Climate Action (SDG 13)**: Promotes understanding of atmospheric dynamics by applying gas laws to global warming and greenhouse gases, fostering a scientific basis for understanding how gas expansion, density, and diffusion affect our changing climate.<br>- **Values Integration (Stewardship, Mindfulness, and Critical Thinking)**: Instills a sense of stewardship for the air we breathe by realizing that invisible gases (pollutants/emissions) have tangible consequences, encouraging mindful decisions regarding carbon footprints and environmental care. |
| F. **Skills** *(21st century skills)* | - Critical Thinking and Problem Solving<br>- Creativity and Innovation<br>- Collaboration |
| Preliminaries | - Prayer<br>- Attendance Checking |

**III. Teaching and Learning Process (Experiences)**

**I. Recapitulation (5 minutes)**

The learners will recall the properties of matter.

- Matter is anything that occupies space, and has mass.

The learners will further elaborate on the properties of solids, liquids, and gas

- Solids
  - Rigid, and can only vibrate in place

- Liquids
    - Fixed volume, takes the shape of the container
- Gas
    - No fixed volume, no fixed shape

**A. Activity (5 minutes)**

**Plan A**

In this activity, the students will arrange their seats in a U-formation to leave space for the center. The students will then be divided into 4 groups. For each round of the activity, 5 members of one group (the presenters) will be designated as the "Gas Particles" and positioned in the center of the open space, while the remaining three groups (non-presenters) will link arms to form a closed circle around them, acting as container walls.

The students representing the gas particles will be instructed to move continuously in straight lines, and will only change direction upon colliding (which will be seen as a high five) with another gas particle, or if they hit the wall.

The students forming the container will be responsible for containing the gas particles, and counting the frequency of collisions they experience.

The first group will show the concept of pressure, the second, temperature, demonstrated with faster/slower walking speeds, the third, volume, demonstrated by the shrinking of the container, and fourth, number of gas particles, in which case, instead of 5 particles in the middle, they start with 7.

**Plan B**

The students will be divided into 2 groups. The groups will be tasked to define two key gas properties, pressure, and temperature, in terms of gas terms. To facilitate this, the teacher will conduct a guided simulation (PhET Gas Properties) projected on the main screen. While the simulation runs, the groups will be given specific observational focal points:

- Group A (Temperature Focus): This group will be instructed to observe the motion of the particles as the group adds heat to the system. They must formulate a definition of Temperature based on the relationship they see between the thermometer readings and the speed of the particles. - Temperature is the average kinetic energy of the particles in a gas system. It also describes the average speed of each particle.

  They are given the following words: Average Kinetic Energy, Speed, Particles. An extra 5 points is awarded if they incorporate the words in the definition that they come up with.

- Group B (Pressure Focus): This group will be instructed to observe the walls of the container, and the number of containers. They must formulate a definition of Pressure based on the interaction they witness between the flying particles and the container walls. - Pressure is the frequency of collisions with the particles and the walls of the container.

They are given the following words: Frequency, Collisions, Walls. An extra 5 points is awarded if they incorporate the words in the definition that they come up with.

After the simulation, the groups will be given two minutes to synthesize their observations and construct a scientific definition.

**B. Analysis (5 minutes)**

**(For Plan A)**

The rest of the group members who were not involved in acting as the gas particles will answer the following guide questions for their group:

Group 1:

1. In this activity, we said "Pressure" is the number of hits you felt. Based on your experience acting as the wall, how would you define *Gas Pressure* in your own words?

Group 2:

2. Compare the slow round to the fast round. In which round did you feel more collisions against your arms?

Group 3:

3. When you stepped forward to make the circle smaller, did the students inside speed up, or did they just run out of space?

Group 4:

4. Think back to Round 1 when we had only 4 students inside. In this round, we started with 7 students immediately. How did the *frequency* of hits feel compared to that first round?

**(For Plan B)**

After the two minutes for synthesis. The groups will have a representative present to the class the definition that they came up with.

The teacher will verify if the required keywords were successfully integrated to award the bonus points. Following this, the teacher will ask the question for group A:

- Did you notice if every single particle was moving at the exact same speed, or were some faster than others? - Some are faster than others. Temperature is the measure of the average speed of the entire system, not individual particles.

The teacher will do the same for the second group, and ask the question:

- If the particles suddenly stopped moving completely, would there still be pressure on the walls? - No. Pressure is caused by the collisions of the particles against the walls of the container.

### C. Abstraction (20 minutes)

The learners will analyze the content presented in the PowerPoint slides, listen to key explanations, and participate by answering questions in-between. They will also take turns reading and interpreting the information on the slides to support understanding. The following information is included in the slides:

a. Introduction to Kinetic Molecular Theory:
   - Kinetic Molecular Theory is a scientific model that explains the *behavior of gases*. It posits that…
   - Gases are composed of molecules. The particles are far apart and occupy almost all the volume of the container.
   - Gas particles are in continuous, random motion.
   - There is negligible attractive or repulsive force between or among gas molecules.
   - Collisions between gas particles are completely elastic.
   - Movement of gas molecules is affected by temperature.

b. Question and Answer: Can you touch and see gas? How can we tell that it is really there and that it takes up space?

   - No. Gases are invisible to the naked eye but they can be observed through four physical properties: *pressure, volume, amount of gas, and temperature*.

     - it takes up space and pushes on things. (e.g. a balloon inflating)
     - It exerts force (e.g. wind blowing on your face)
     - It has mass/weight (e.g. If you weigh an empty balloon vs. a full balloon on a sensitive scale, the full one is heavier.)
     - It heats up (e.g. balloon expanding over high heat)

c. Properties That Described a Gas: Volume (V)

   - Volume (denoted by the symbol V) is described as the space occupied by a given matter.
   - This means that gases do not have definite shape but occupy only the space of the container.
   - Is usually measured in liters (L) or milliliters (mL), and in cubic units.

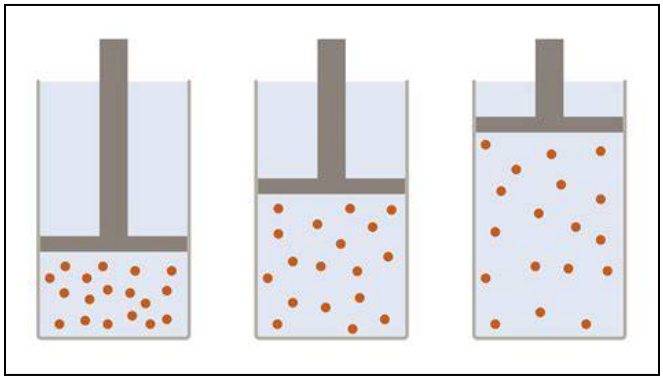
d. Properties That Describe a Gas: Amount of Gas (n)
   - The number of gas particles in a container, usually measured in moles (n).
   - 1 mole = $6.02214076 \times 10^{23}$ elementary units.

e. Properties That Described a Gas: Pressure (P)

   - Pressure (denoted by the symbol P) is described as a force acting on a specific area.

- It is the collisions of the gas molecules with the walls of the container.
- The magnitude of the pressure is related to how hard and how often the molecules strike the wall.

  - More collisions = greater pressure
  - Stronger collisions = greater pressure



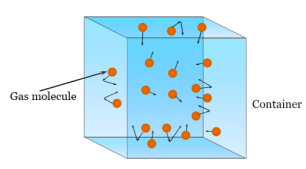
- Pressure is measured using barometers, and has several units:

| Unit | Abbreviation | Unit Equivalent to 1 atm |
|---|---|---|
| atmosphere | atm | 1 atm (exact) |
| millimeters of Hg | mmHg | 760 mmHg (exact) |
| torr | Torr | 760 Torr (exact) |
| inches of Hg | inHg | 29.9 inHg |
| pounds per square inch | $lb/in.^2$ (psi) | 14.7 $lb/in.^2$ |
| pascal | Pa | 101 325 Pa |
| kilopascal | kPa | 101.325 kPa |

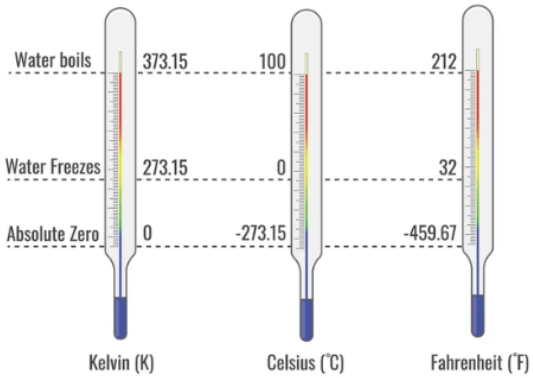
- Atmospheric Pressure is the pressure exerted by the Earth's atmosphere and can be measured using a barometer.
- Evangelista Torricelli invented the barometer in the 17th century.
- A Torricelli barometer is a glass tube, closed at one end, filled with mercury, about 760 mm long.

f. Properties That Described a Gas: Temperature (T)

- Temperature (T) is how hot or cold an object is and can be measured with a thermometer.
  - It is the average kinetic energy of all the constituents of a system.
  - It describes the speed of the gas particles wherein higher temperature = higher kinetic energy = faster particles.



- Gas temperature is measured in Kelvin (K) for calculations.
  - Kelvin starts at absolute zero (0 K = –273.15°C), the lowest possible temperature where particles stop moving.




## TEMPERATURE CONVERSIONS

| To Convert From: | Use this equation: |
|---|---|
| ° C → ° F | ° F = (9/5 × ° C) + 32 |
| ° F → ° C | ° C = 5/9 (° F – 32) |
| ° C → K | K = ° C + 273 |
| K → ° C | ° C = K - 273 |

**D. Application (10 minutes)**

The 2 groups will be tasked to explain 1 situation each.

Group 1: A truck driver fills his tires to the maximum limit early in the morning. He drives for 4 hours on a very hot asphalt highway. Suddenly, one of his tires explodes.

Why did the tire burst even though he didn't add any more air?

Group 2: During a PE game, someone accidentally steps on a ping-pong ball, leaving a large dent in it. The teacher doesn't throw it away. Instead, she drops the dented ball into a cup of boiling water. After a few seconds, the dent pops out, and the ball is round again.

How did the hot water fix the ball without anyone touching it?

| **IV. Learning Assessment** |
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| No assessment task will be given on this day. |
| **V. Assignment** |
| Directions: On a ½ crosswise paper, research Boyle's Law and Charles' Law. For each law, write:<br>1. Who discovered it<br>2. What the law states<br>3. One real-life example<br><br>Be ready to share your findings in the next class. |